

EDUCATION BACKGROUND

Xiamen University Malaysia

Sep. 2022-Jun. 2026

BSc (Honor) in Marine Environmental Chemistry

- **GPA:** 3.67/4.0

- **Core Courses:** Remote Sensing and GIS, Environmental Monitoring, Environmental Pollution Control Technology, Instrumental Analysis, Environmental Modeling, Environmental Management, Descriptive Oceanography, etc.

- **Honors & Awards:** Dean's List (2025), Distinction Award (three times, 2023&2024&2025)

Leibniz Centre for Tropical Marine Research, ZMT

Jul. 2025-Aug. 2025

IOI Online Summer School 2025

- Completed intensive training on transdisciplinary coastal and marine research, mastering core methodologies including research co-design, complex system analysis, and science-policy communication

RESEARCH & PROJECTS

Nearshore Erosion Dynamics in an Estuarine Channel-Shoal System

Sep. 2025-Present

Research Intern, State Key Lab of Estuarine and Coastal Research, East China Normal University

- **Data Mining:** Processed 10+ years of shoreline and bathymetric datasets via Python-based Coastsat model and spatial analytics, quantified erosion hotspots with 1-4 m/yr retreat; correlated peaks with storms/precipitation via time-series mining
- **Model Development:** Developed and calibrated a Delft3D morphodynamic model achieving ~5% water level accuracy, capturing wave-tide coupling and monsoon-based sediment transportation
- **Driver Diagnosis:** Used Pandas/NumPy for feature engineering, integrated model outputs with field data to diagnose drivers; quantified strategies-30-40% erosion reduction via nourishment, 0.5-1 m/yr downdrift erosion from hard structures

Generating Impact Through Transdisciplinary Research: Co-Designing Solutions for Complex Marine and Coastal Challenges

Jul. 2025-Aug. 2025

Group Member, Leibniz Centre for Tropical Marine Research, ZMT

- **Vulnerability Assessment:** Synthesized high-resolution (100-250 m) regional datasets using Python (Pandas, GeoPandas) for spatial data integration, quantitatively evaluating ecological and social vulnerability of Jakarta Bay coastal communities
- **Systems Mapping & Leverage Identification:** Collaborated with an international team to develop a transdisciplinary systems map, encoding 12 key drivers (groundwater extraction, mangrove loss, flooding feedbacks) and their interdependencies; applied network analysis to identify critical leverage points for coastal resilience planning
- **Intervention Impact Evaluation:** Compared three intervention pathways (mangrove rehabilitation, hybrid grey-green barriers, groundwater regulation) using scenario modeling; quantified 5-year/10-year impacts on risk exposure reduction and community feasibility via statistical analysis, providing data-informed policy recommendations

2024 Marine Silk Road Scientific Training Program, R/V TTK (Second Leg)

Aug. 2024-Sep. 2024

Nutrient Team Leader, South China Sea Research Cruise

- **Sampling Design:** Led nutrient-analysis operations across four hydrographic stations, designing water-column sampling based on cruise objectives and real-time CTD profiles
- **Onboard Measurement & Calibration:** Conducted in-situ measurements using CTD rosette, nutrient analyzer, TA/DO instruments; implemented strict clean sampling standards and systematic calibration to ensure data accuracy
- **Data QA/QC & Validation:** Established standardized sample processing workflows; performed rigorous QA/QC procedures to validate nutrient and carbonate-system datasets, eliminating outliers and ensuring data reliability
- **Data Consolidation & Support:** Compiled cruise logs and integrated field data to support the team's biogeochemical data products, demonstrating strong field operational capability and cross-team coordination under shipboard constraints

Algal Pigment and Metabolite Analysis Using HPLC

_Jan. 2024

Research Assistant, Hong Kong University of Science and Technology

- **HPLC-Based Analysis:** Conducted pigment and metabolite quantification on cultured algal samples via HPLC, executing solvent extraction, chromatographic separation, and peak interpretation using Agilent ChemStation software
- **Data Traceability & Documentation:** Maintained comprehensive experimental records (sample IDs, extraction parameters, instrument settings) to ensure full data traceability, adhering to research-grade laboratory protocols
- **QA/QC Optimization:** Refined QA/QC workflows (blank sample runs, standard curve calibration) and added contamination-check steps, improving pigment quantification consistency by 18% while meeting rigorous laboratory standards

EXTRACURRICULAR ACTIVITIES

Xiamen University 70.8 Media Lab

-Apr. 2025-Nov. 2025

Science Writer & Editor, Xiamen University

- **Outreach Article Production:** Produced 12+ accessible marine science articles covering ocean ecology, environmental processes, and fundamental scientific concepts; translated technical content (e.g., biogeochemical cycles, marine pollution monitoring) into engaging narratives for non-specialist audiences
- **Content Quality Assurance:** Reviewed and refined 15+ manuscripts for scientific accuracy and readability, conducting literature checks, verifying data sources, and preparing 8+ supporting visuals/infographics to enhance content accessibility

"Jia Geng" Marine Science Exhibition

Aug. 2024

Science Outreach Volunteer, Hong Kong & Kuala Lumpur

- **Marine Science Dissemination:** Presented marine science concepts and research vessel operations to 200+ public visitors (including students and families), clearly explaining oceanographic missions, shipboard instruments (e.g., CTD rosettes, sediment corers), and their applications in environmental research
- **Cross-Institutional Engagement:** Facilitated interactive exchanges with 50+ Hong Kong Polytechnic University participants during the joint outreach event, coordinating discussions on marine conservation and science communication

ADDITIONAL INFORMATION

- **Languages:** Chinese (native), English (TOEFL 101, GRE 315)

- **Skills:** Python, MATLAB, R, C, Delft3D, Ocean Data View (ODV), GIS & Remote Sensing tools, HPLC operation

- **Hobbies:** Swimming, Badminton, Kendo (9th-kyu), Recreational Diving